Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

To implement and compare the performance of:

- **Model A:** Decision Tree classifier
- **Model B:** Random Forest ensemble model (extension of Decision Trees)

Students must study the impact of hyperparameters on overfitting and generalization, perform hyperparameter tuning using 5-Fold Cross-Validation, and compare the performance between single-tree and ensemble-based models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: https://archive.ics.uci.edu

Theory

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures to form decision rules.

Key Concepts:

- Gini Index and Entropy
- Tree depth and node splitting
- Overfitting in deep trees

Limitation: Decision Trees are prone to high variance and overfitting.

Random Forest Classifier

Random Forest is an ensemble learning technique that combines multiple decision trees trained on bootstrapped samples.

Key Ideas:

- Bootstrap aggregation (bagging)
- Random feature selection
- Majority voting

Advantage: Reduces variance and improves generalization compared to a single Decision Tree.

Implementation Summary

The dataset was loaded and the target labels were encoded. Exploratory Data Analysis (EDA) was performed to observe class distribution and feature correlations. The data was then split into an 80–20 training and testing set.

Both Decision Tree and Random Forest classifiers were trained, and optimal hyperparameters were selected using 5-Fold Cross-Validation (GridSearchCV). Finally, the models were evaluated and compared using multiple performance metrics.

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation (Top 5 Results)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	3	94.29	92.32
entropy	5	94.29	92.06
entropy	5	94.29	92.06
entropy	10	94.29	92.06
entropy	None	94.29	92.06

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	10	sqrt	96.48	95.15
50	20	sqrt	96.48	95.15
50	None	sqrt	96.48	95.15
200	None	sqrt	96.26	94.88
100	None	log2	96.26	94.92

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9780	0.9121	0.9451	0.9451	0.9341	0.9429
Random Forest	0.9780	0.9560	0.9670	0.9780	0.9451	0.9648

Hyperparameter Analysis

In this experiment, optimal model configurations were identified using 5-Fold Cross-Validation. The hyperparameter search space was designed to balance the bias-variance tradeoff, with a particular focus on mitigating the overfitting tendencies of Decision Trees.

1. Decision Tree Parameter Analysis

The Decision Tree tuning focused on parameters that control node splitting and tree growth:

- **Criterion (gini vs. entropy):** This parameter determines the function used to measure the quality of a split. While Gini impurity is computationally faster, Entropy can sometimes produce more balanced trees by maximizing information gain. The observed differences in performance were minimal for this dataset.
- **Max Depth (`max_depth`):** This is the most critical hyperparameter for controlling overfitting. Allowing unrestricted depth (None) often leads to memorization of training data, reducing generalization. Limiting the depth (e.g., 3 or 5) acts as pre-pruning and helps capture only the most significant patterns.
- **Minimum Samples (`min_samples_split`, `min_samples_leaf`):** Increasing these values enforces stricter conditions for node splitting and leaf formation. This acts as regularization, resulting in smoother decision boundaries and reduced variance.

2. Random Forest Parameter Analysis

The Random Forest tuning focused on ensemble-level parameters and constraints on individual trees:

- **Number of Estimators (`n_estimators`):** This defines the number of trees in the forest. Increasing this value improves stability and reduces variance, but performance tends to plateau beyond a certain point (typically around 100–200 trees).
- **Max Features (`max_features`):** Limiting the number of features considered at each split (e.g., `sqrt`, `log2`) enforces randomness among trees. This de-correlation is key to reducing variance in the ensemble.
- **Bootstrap:** Enabling bootstrapping ensures that each tree is trained on a randomly sampled subset of the dataset with replacement. This increases robustness to noise and outliers.
- **Max Depth (`max_depth`):** Unlike single Decision Trees, Random Forests can tolerate deeper trees. Since predictions are averaged across multiple trees, the overfitting of individual trees is mitigated, allowing the model to learn more complex patterns.

Evaluation Metrics

Performance Evaluation

Table 4: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.9649	1.0000	0.9070	0.9512
Random Forest	0.9561	0.9524	0.9302	0.9412

Table 5: Confusion Matrix - Decision Tree

	Predicted 0	Predicted 1
Actual 0	71	0
Actual 1	4	39

Table 6: Confusion Matrix - Random Forest

	Predicted 0	Predicted 1
Actual 0	69	2
Actual 1	3	40

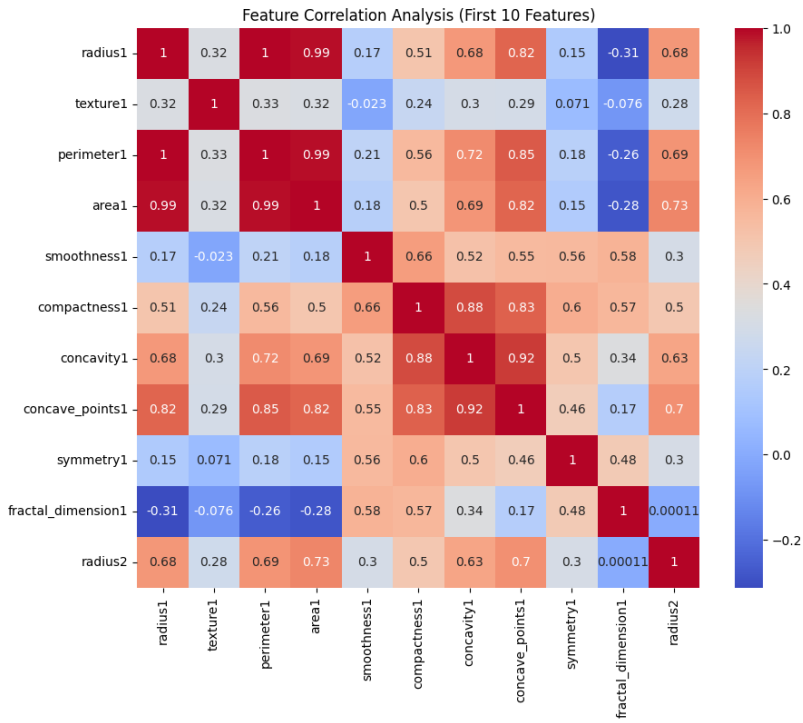

Figure 1: Feature Correlation Analysis (First 10 Features)

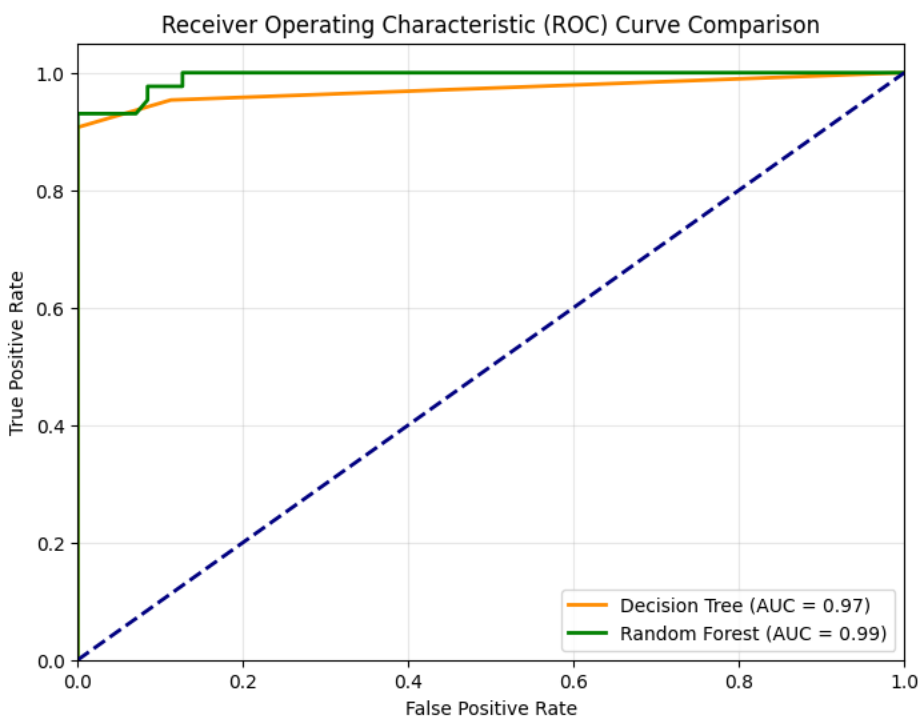

Figure 2: ROC Curve

Observations

- **How does tree depth affect overfitting in Decision Trees?**
  Increasing tree depth allows the model to learn overly complex rules that perfectly classify the training data, capturing noise and outliers. This leads to high variance and severe overfitting, reducing performance on unseen test data.

- **Which hyperparameter had the greatest impact on performance?**
  For the Decision Tree, `max_depth` and `min_samples_split` had the greatest impact by controlling tree complexity. For the Random Forest, `n_estimators` (number of trees) significantly improved stability and reduced variance.

- **How does Random Forest improve generalization?**
  Random Forest improves generalization using Bootstrap Aggregation (bagging), where multiple trees are trained on different subsets of the data. Additionally, random feature selection at each split ensures the trees are less correlated. Averaging these trees reduces overall variance.

- **Did ensemble learning always improve performance? Why or why not?**
  Ensemble learning generally improved stability and test-set performance by reducing variance. However, if the individual trees are too weak (e.g., very shallow trees) or if the dataset is simple, the improvement over a well-tuned single tree may not be very significant.

Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset